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DEVICE FOR TRAPPING ONE OR MORE PARTICLES AND ASSOCIATED METHOD AND SYSTEM

Abstract

A device for trapping one or more particles includes first and second RF electrodes and DC electrodes. The first and second RF electrodes extend in a first direction and both have opposite first and second sides, with the first sides facing towards one another. In a center region of the first and second RF electrodes, the first sides have a first distance with respect to each other. Also in the center region, the second sides have a second distance with respect to each other. In an ending region of the first and second RF electrodes: the first sides have a distance that is greater than the first distance; and/or the second sides have a distance that is less than the second distance.

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Background/Summary

BACKGROUND

[0001] Trapped particles (such as electrons, ions or molecules) are a promising candidate for use as qubits (quantum bits) in quantum computers. Moreover, trapped particles (such as electrons, ions or molecules) can be used in metrology (e.g., in atomic clocks).

[0002] Felix Stopp et al (2021) arXiv:2108.06948v1 [quant-ph] discloses that an ion trap is used where ions are extracted and sent to free space. The ion trajectories are reflected and the ions are captured entering from free space in the ion trap where they started. Both, in order to send the ions to free space and in order to capture the ions from free space, the radio frequency (RF) signal of the ion trap is ramped (e.g., by increasing/decreasing the amplitude of the applied RF voltage over a given period of time).

[0003] In practical applications, the exact tuning of the ramping of the RF signal may be challenging. Therefore, it may be desirable to provide an improved trapping device that may allow sending one or more particles (such as ions) away from the trapping device and/or receiving one or more particles (such as ions). It may also be desirable to provide a system and a method that may use at least some of the above-described principles.

SUMMARY

[0004] According to embodiments, a device for trapping one or more particles comprises a first radio-frequency (RF) electrode, a second RF electrode, and a plurality of direct current (DC) electrodes. The first RF electrode and the second RF electrode may extend in a first direction. The first RF electrode may comprise a first side and a second side. The second side of the first RF electrode may be arranged opposite to the first side of the first RF electrode. The second RF electrode may comprise a first side and a second side. The second side of the second RF electrodes may be arranged opposite the first side of the second RF electrode. The first side of the first RF electrode may face towards the first side of the second RF electrode. In a center region of the first RF electrode and the second RF electrode, the first side of the first RF electrode and the first side of the second RF electrode may have a first distance d.sub.1,c with respect to each other. In the center region of the first RF electrode and the second RF electrode, the second side of the first RF electrode and the second side of the second RF electrode may have a second distance d.sub.2,c with respect to each other. In an ending region of the first RF electrode and the second RF electrode: the first side of the first RF electrode and the first side of the second RF electrode have a distance d.sub.1 that is greater than the first distance d.sub.1,c; and/or the second side of the first RF electrode and the second side of the second RF electrode have a distance de that is less than the second distance d.sub.2,c.

[0005] According to embodiments, a system comprises a first device for trapping one or more particles, and a second device for trapping one or more particles. The first device may be configured to trap at least one particle provided by a particle source. The second device may be configured to receive the at least one particle from the first device, and to perform one or more quantum gate operations on the at least one particle.

[0006] According to embodiments, a method of providing one or more particles to a device for trapping one or more particles, the method comprises trapping, by a first device, at least one particle provided by a particle source. The method may further comprise providing, by the first device, the at least one particle to a second device. The method may further comprise receiving, by the second device, the at least one particle. The method may further comprise performing, by the second device, one or more quantum gate operations on the at least one particle. Those skilled in the art will recognize additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar or identical elements. The elements of the drawings are not necessarily to scale relative to each other. The features of the various illustrated examples can be combined unless they exclude each other and/or can be selectively omitted if not described to be necessarily required.

[0008] FIG. **1**A illustrates a top view of a device for trapping one or more particles (such as ions) according to an example of the present disclosure.

[0009] FIG. **1**B illustrates a top view of a device for trapping one or more particles (such as ions) according to another example of the present disclosure.

[0010] FIG. **1**C illustrates a top view of a device for trapping one or more particles (such as ions) according to yet another example of the present disclosure.

[0011] FIG. **2**A illustrates a cross sectional view of a device for trapping one or more particles (such as ions) according to an example of the present disclosure.

[0012] FIG. **2**B illustrates a cross sectional view of a device for trapping one or more particles (such as ions) according to an example of the present disclosure.

[0013] FIG. **2**C illustrates a cross sectional view of a device for trapping one or more particles (such as ions) according to another example of the present disclosure.

[0014] FIG. **3** illustrates a top view of a device for trapping one or more particles (such as ions) according to an example of the present disclosure.

[0015] FIG. **4** illustrates a top view of a device for trapping one or more particles (such as ions) according to an example of the present disclosure.

[0016] FIG. **5** illustrates a top view of a device for trapping one or more particles (such as ions) according to an example of the present disclosure.

[0017] FIG. **6** illustrates a top view of a device for trapping one or more particles (such as ions) according to an example of the present disclosure.

[0018] FIG. **7** illustrates a top view of a device for trapping one or more particles (such as ions) according to an example of the present disclosure.

[0019] FIG. **8** illustrates a system including a first device for trapping one or more particles (such as ions) and a second device for trapping one or more particles according to an example of the present disclosure.

[0020] FIG. **9** illustrates a method for providing one or more particles (such as ions) to a device for trapping one or more particles.

[0021] Further examples are explained below with reference to the drawings. The drawings serve to illustrate certain principles, so that only aspects necessary for understanding these principles are illustrated in the drawings. Additional elements serving different functionality may be present in the illustrated devices, methods and systems, but may not be illustrated in the drawings.

DETAILED DESCRIPTION

[0022] The devices described herein may be configured to trap one or more particles (such as ions, molecules or electrons) and control the trapped particles. In particular, the trapped particles may be physically grouped in particle chains (or strings or crystals). A particle chain may include one or multiple particles of the same or different species, wherein each of the particles (e.g., ions) may represent a physical qubit. In some examples, the devices described herein may be used for quantum computing, but are not restricted thereto. Trapped ions are one of the most promising candidates for being used as qubits in quantum computers, since they can be trapped with long lifetimes in a scalable array by means of electromagnetic fields. Other applications may be in the field of atomic clocks.

[0023] The trapped particles may be shuttled (or transported) along shuttling paths of the device. The shuttling paths may include straight sequences, but also junctions, such as X- junctions and/or T-junctions. For example, the shuttling paths may extend above multiple electrodes of the trapping device. Time-dependent electric fields may be used for shuttling the particles along the shuttling paths. A shuttling of the particles may be controlled by electric voltages applied to the electrodes of the device. In particular, the particles may be moved along shuttling paths by means of alternating current (AC) and direct current (DC) voltages that may be separately coupled to specific electrodes of the device. For example, the electrodes may include one or more radio-frequency (RF) electrodes for RF trapping and a plurality of DC electrodes for static electric-field trapping and/or for moving the particles within the trapping device. Trapping devices as described herein may be configured to trap a plurality of particles that may be individually addressable and movable by appropriately controlling the electric voltages of the electrodes.

[0024] In a specific but non-limiting example, the trapping devices described herein may correspond to or may include a surface trapping device (such as a Paul-trap) where all electrodes (i.e. the DC electrodes and the RF electrodes) may be arranged in a same single plane (e.g., in a structured electrode layer arranged on a substrate). The particles may be trapped and shuttled above this single plane. However, it is to be understood that the concepts described herein are not restricted to surface trapping devices. In further examples, devices for controlling trapped particles in accordance with the disclosure may also be based on three-dimensional (3D) trap geometries.

[0025] FIGS. **1**A-**1**C illustrate typical layouts of a device **100**A, **100**B, **100**C for trapping one or more particles. The drawings serve to illustrate certain principles, so that only aspects necessary for understanding these principles are illustrated. For example, the devices **100**A, **100**B, **100**C may include additional DC electrodes, e.g., for compensating straight fields (e.g., connected to a ground potential or to another compensation potential) and/or for shielding that are not illustrated in the drawings.

[0026] The device **100**A, **100**B, **100**C comprises a first radio-frequency (RF) electrode **110**, a second RF electrode **120**, and a plurality of direct current (DC) electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**. These electrodes may be arranged above a substrate **160**.

[0027] The first RF electrode **110** may extend in a first direction x and may comprise a first

[0028] side **112**, a second side **114**, and a first end **116**. The first RF electrode **110** may further comprise a second end further downwards in the x-direction which is not shown in FIGS. **1**A-**1**C. The first side **112** and the second side **114** of the first RF electrode **110** are spaced apart from each other along a second direction y (which may be orthogonal to the first direction x). The second side **114** of the first RF electrode **110** is arranged opposite to the first side **112** of the first RF electrode **110**. Typically, the first end **116** and also the second end of the first RF electrode has a straight shape and is arranged with a 90° angle with respect to the first side **112** and the second side **114** of the first RF electrode **110**.

[0029] The first direction x and the second direction y may be parallel to a first major surface of substrate **160** and may be referred to as lateral directions. Typically, particles may be trapped in a region that is distanced from the first RF electrode **110** and the second RF electrode **120** in a third direction z (which may be orthogonal to the first direction x and the second direction y). The third direction may be orthogonal to a first major surface of the substrate **160** and may be referred to as a vertical direction. The second RF electrode **120** may extend in the first direction x and may comprise a first side **122**, a second side **124**, and a first end **126**. The second RF electrode **120** may further comprise a second end further downwards in the x-direction which is not shown in FIGS. **1**A-**1**C. The first side **122** and the second side **124** of the second RF electrode **120** are spaced apart from each other along the second direction y. The second side **124** of the second RF electrode **120** is arranged opposite to the first side **122** of the second RF electrode **120**. Typically, the first end **126** and also the second end of the second RF electrode **120** has a straight shape and is arranged with a 90° angle with respect to the first side **122** and the second side **124** of the second RF electrode **120**.

[0030] In the device **100**A, **100**B, **100**C as illustrated in FIGS. **1**A-**1**C, the first side **112** of the first RF electrode **110** faces towards the first side **122** of the second RF electrode **120**. In other words, the first side **112** of the first RF electrode and the first side **122** of the second electrode may be arranged closer to each other as compared to the second sides **114** and **124** of the first and second RF electrodes **110** and **120**.

[0031] The device **100**A, **100**B, **100**C comprises a center region **170** and an ending region **180** of the first RF electrode **110** and the second RF electrode **120**. In the center region **170** of the first RF electrode **110** and the second RF electrode **120**, the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120** may have a first distance d.sub.1,c with respect to each other. The first distance d.sub.1,c may be measured along the second direction y (e.g., in case of a surface trapping device as illustrated with regard to FIG. **2**A or with regard to a 3D trapping device as illustrated with regard to FIG. **2**B). In other examples, the first distance d.sub.1,c may be measured along a direction (which may be different from the second direction y) that lies in a plane formed by the second direction y and the third direction z (as illustrated with regard to FIG. **2**C) and that is orthogonal to the first direction x.

[0032] In the center region **170** of the first RF electrode **110** and the second RF electrode **120**, the second side **112** of the first RF electrode **110** and the second side **122** of the second RF electrode **120** have a second distance d.sub.2,c with respect to each other. The second distance d.sub.2,c may be measured along the second direction y (e.g., in case of a surface trapping device as illustrated with regard to FIG. **2**A or with regard to a 3D trapping device as illustrated with regard to FIG. **2**B). In other examples, the first distance d.sub.1,c may be measured along a direction (which may be different from the second direction y) that lies in a plane formed by the second direction and the third direction (as illustrated with regard to FIG. **2**C) and that is orthogonal to the first direction x. The second distance d.sub.2,c is greater than the first distance d.sub.1,c.

[0033] In the illustrated examples of FIGS. **1**A-**1**C, the first distance d.sub.1,c is substantially constant in the center region **170** and in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**. Similarly, the second distance d.sub.2,c is substantially constant in the center region **170** and in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**. In the context of the present disclosure a value being substantially constant or two values being substantially the same shall mean that the value is constant or the two values are the same within typical processing variations. In an example, the first distance d.sub.1,c may be in a range between 10 μm and 1000 μm, such as between 100 μm and 250 μm (such as 160 μm) and the second distance d.sub.2,c may be in a range between, 50 μm and 5000 μm, such as between 500 μm and 750 μm (such as 660 μm). However, other values are possible and these exemplary values should not be construed limiting.

[0034] The first RF electrode **110** of device **100**A, **100**B, **100**C may have a width that may be defined as a third distance d.sub.3,c. between the first side **112** of the first RF electrode **110** and the second side **114** of the first RF electrode **110**. The third distance d.sub.3,c. may be measured along the second direction y. The second RF electrode **110** of device **100**A, **100**B, **100**C may have a width that may be defined as a fourth distance d.sub.4,c. between the first side **122** of the second RF electrode **120** and the second side **124** of the second RF electrode **110**. The fourth distance d.sub.4,c. may be measured along the second direction y. As illustrated with respect to FIGS. **1**A-

1C, device **100**A, **100**B, **100**C the third distance d.sub.3,c is substantially constant and the fourth distance d.sub.4,c is substantially constant in the center region (**170**) and in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**. The third distance d.sub.3,c and the fourth distance d.sub.4,c may be measured along the second direction y.
[0035] In some examples the third distance d.sub.3,c may be substantially equal to the fourth distance d.sub.4,c. In other embodiments, the third distance d.sub.3,c may differ from the fourth distance d.sub.4,c. In an example, the third distance d.sub.3,c may be in a range between 20 μm and 3 mm, such as between 100 μm and 750 μm (such as 250 μm) and the fourth distance d.sub.4,c may be in a range between, 100 μm and 3 mm, such as between 100 μm and 750 μm (such as 250 μm). However, other values are possible and these exemplary values should not be construed limiting.
[0036] The first and second RF electrodes **110** and **120** of devices **100**A, **100**B, **100**C as exemplarily illustrated in FIGS. **1**A-**1**C are essentially the same. The devices **100**A, **100**B, and **100**C only differ with respect to the arrangement of the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**. Generally, the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** are arranged in proximity (such as adjacent) to the first RF electrode **110** and the second RF electrode **120**. For example, the first RF electrode **110**, the second RF electrode **120** and the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** may be part of a (common) metallization structure (e.g., obtained by structuring one or more metallization layers) arranged on substrate **160**.
[0037] As shown in FIGS. **1**A-**1**C, the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** may be arranged adjacent to the first RF electrode **110** and the second RF electrode **120** in the center region **170** and optionally also in the ending region **180**.
[0038] Referring now to FIGS. **1**A and **1**B, the device **100**A, **100**B may have a first set **130**.sub.1, . . . , **130**.sub.n of the plurality of DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** being arranged adjacent the second side **114** of the first RF electrode **110** and a second set **140**.sub.1, . . . , **140**.sub.n of the plurality of DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** being arranged adjacent the second side **124** of the second RF electrode **120**.
[0039] In addition to that, and as shown in FIGS. **1**A and **1**B, at least a further DC electrode **150** of the plurality of DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** may be arranged between the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120**. This further DC electrode **150** may be a single electrode (as exemplarily illustrated in FIG. **1**A) or comprise a plurality of segments (as exemplarily illustrated in FIG. **1**B).
[0040] Another example is exemplarily illustrated with respect to FIG. **1**C, where the first set **130**.sub.1, . . . , **130**.sub.n of the plurality of DC electrodes and the second set **140**.sub.1, . . . , **140**.sub.n of the plurality of DC electrodes is not present. In the exemplary embodiments illustrated by FIGS. **1**A and **1**B, the first set **130**.sub.1, . . . , **130**.sub.n of the plurality of DC electrodes and the second set **140**.sub.1, . . . , **140**.sub.n of the plurality of DC electrodes may be used for providing a trapping potential and the at least one further DC electrode **150** may be used for one or more of shielding, straight field compensation, and enhancing the trapping potential. The device **100**C, as illustrated in FIG. **1**C comprises a plurality (e.g., three or more) of the at least one further DC electrode **150** that is arranged between the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120**. In the exemplary embodiment shown in FIG. **1**C, the plurality of further DC electrodes **150** is configured to provide a trapping potential for trapping one or more ions. The exemplary device **100**C shown in FIG. **1**C comprises at least seven DC electrodes **150**. However, the present disclosure is not limited to seven DC electrodes **150**, but more or less electrodes **150** can be present. In an example, only three DC electrodes **150** could be present.
[0041] Referring back to any of FIGS. **1**A-**1**C, substrate **160** may have an upper surface (also

referred to as first major surface) on which the first RF electrode **110**, the second RF electrode **120**, and the plurality of DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** are arranged. In examples, the upper surface of substrate **160** may be substantially planar. In embodiments, the substrate **160** may comprise one or more of semiconductor material (such as Si, SiC and GaN or similar materials), fused silica, quartz glass, and sapphire. While not limited to these values, the substrate **160** may have a thickness of 400 μm to 1 mm (such as 725 μm or 750 μm for example).

[0042] The first RF electrode **110**, the second RF electrode **120**, and the plurality of DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** may be connected to signal lines (not shown in FIGS. **1**A-**1**C) for providing respective signals to these electrodes. For example, the signal lines can be conductors arranged on the surface of substrate **160**. In addition to that or as an alternative, signal lines can be routed in a further structured metal layer that is arranged between the RF and/or DC electrodes **110**, **120**, **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** and the substrate **160**. Yet as a further alternative, signal lines may reach through the substrate **160** and connect the RF and/or DC electrodes **110**, **120**, **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** to a metallization structure on the opposing side of the substrate **160**. The signal lines may be connected to respective signal generators (such as digital-to-analog converters in case of the

[0043] DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** or a resonator in case of the first and second RF electrodes **110**, **120**) that provide the respective signals to the electrodes. The respective signal generators may be connected via one or more intervening circuit elements, such as capacitors or other filters.

[0044] The first RF electrode **110**, the second RF electrode **120**, and the plurality of DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** are configured to create a trapping potential (such as an electromagnetic field) for trapping one or more particles (such as such as ions, molecules or electrons). For example, the plurality of DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** may create a potential (e.g., a static electric field) that confines the one or more particles in the first direction x, and the first RF electrode **110** and the second RF electrode **120** may create a potential (e.g., an RF electromagnetic field) that confines the one or more particles in the second direction y and in the third direction z (which is orthogonal to the first and the second directions). This may result in a potential (e.g., an electromagnetic field) that traps the one or more particles in all three directions in space.

[0045] Typically, the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** are further (such as in addition to trapping) configured to move (such as shuttle) the one or more particles from a first position of the device **100**A, **100**B, **100**C to a second position of the device **100**A, **100**B, **100**C by having respective voltage signals being applied on the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**.

[0046] In practical applications, the signal being applied to the first RF electrode **110** and the second RF electrode **120** may be the same. The signal being applied to the first RF electrode **110** and the second RF electrode **120** typically comprises a constant amplitude A and a constant frequency w and can be denoted as V.sub.RF=A×sin ωt.

[0047] The signals being applied to the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** may vary in time depending on the application. For example, if one or more particles shall be trapped at a position of the device **100**A, **100**B, **100**C, one or a pair of the DC electrodes **130**.sub.1, **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** may provide a lower potential than the remainder of the DC electrodes.

[0048] In the center region **170**, the potential that is created by the first RF electrode **110** and the second RF electrode **120** may have a contribution to the first direction x that is neglectable when being compared to the potential in the first direction x that is being created by the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**. In the ending region **180** of device

100A, **100**B, **100**C the potential that is created by the first RF electrode **110** and the second RF electrode **120** may have a contribution to the first direction x that is not neglectable when being compared to the potential in the first direction that is being created by the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**. In other words, the first RF electrode **110** and the second RF electrode **120** may create a potential barrier in the first direction x in the ending region **180**. This may prevent the one or more particles that are trapped by device **100**A, **100**B, **100**C from leaving the trapping device (e.g., this may slow down and stop one or more particles moving along the first direction x towards the ends **116** and **126** of the first and second RF electrodes **110** and **120**.)

[0049] Referring now to FIGS. **2**A-**2**C, where same reference signs refer to the same or similar elements as in FIGS. **1**A-**1**C. The device **100**A, **100**B, **100**C as illustrated in FIGS. **1**A-**1**C can comprise a single substrate **160**, as exemplarily illustrated in FIG. **2**A (also sometimes referred to as **2**D trapping device **200**A or surface trapping device **200**A). The device **200**A exemplarily illustrated in FIG. **2**A may be a cross-section along the second and third directions y, z (such as the y-z plane) of device **100**A and **100**B shown in FIG. **1**A and **1**B. As discussed with regard to FIG. **1**C further above, some of the DC electrodes of the device **200**A (such as DC electrodes **130** and **140** may be optional (not shown in FIG. **2**A). Again, the devices **200**A, **200**B, **200**C may include additional DC electrodes, e.g., for compensating straight fields (e.g., connected to a ground potential or to another compensation potential) and/or for shielding that are not illustrated in the drawings.

[0050] Examples of 3D trapping devices **200**B, **200**C are exemplarily illustrated in FIGS. **2**B and **2**C, where the trapping device **200**B, **200**C comprises a second substrate **162** (in addition to the first substrate **160**). The second substrate **162** is arranged with a distance to the first substrate **160** along the third direction z. For example, a particle (such as an ion) may be trapped between the first substrate **160** and the second substrate **162**.

[0051] In the example shown in FIG. **2**B, additional DC electrodes **132** and **142** may be arranged on the second substrate **162** such that a surface of the additional DC electrodes **132** and **142** faces towards a surface of the first RF electrode **110**, the second RF electrode **120**, and the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150** that are arranged on the first substrate **160**. The additional DC electrodes **132** and **142** on the second substrate **162** may be fabricated in a similar fashion as the electrodes on the first substrate **160** (e.g., by structuring one or more metal layers on the substrate). In the example shown in FIG. **2**B, the part of device **200**B that is arranged on the first substrate **160** can be the same or similar as the previously discussed devices **100**A, **100**B, **100**C, **200**A. In addition to that, when the additional electrodes **132** and **142** (which each can comprise a plurality of electrode segments) are present, the remaining DC electrodes **130**, **140**, **150** may be omitted in some examples.

[0052] In the exemplary device **200**A and **200**B shown in FIG. **2**A and **2**B, the first RF electrode **110** and the second electrode **120** are arranged on a first substrate **160** such that the first RF electrode and the second RF electrode are arranged within a common plane (which would be the x-y plane illustrated in the examples of FIGS. **2**A and **2**B.

[0053] Referring now to FIG. **2**C, where an exemplary device **200**C is shown, where the first RF electrode **110** is arranged on the first substrate **160** and the second RF electrode **120** is arranged on the second substrate **162** such that the first RF electrode **110** and the second RF electrode **120** are arranged in two separate planes. The two separate planes may be parallel with respect to each other and spaced apart from each other in the third direction z. In this example, the first set of DC electrodes **130** and the at least one further DC electrode **150** is arranged on the first substrate **160** and the second set of DC electrodes **140** is arranged on the second substrate, as can be seen in FIG. **2**C. It is worth mentioning that the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120** are still consider as facing each other within the context of this disclosure (despite the first and second RF electrodes **110** and **120** laying in different planes).

Again, the further DC electrode **150** is optional as described further above.
[0054] The layout of a device **100**A, **100**B, **100**C, **200**A, **200**B, **200**C for trapping one or more particles as shown with respect to FIGS. **1**A-**2**C can be combined in many ways, and the disclosure of the present disclosure as set forth herein shall not be limited to any specific ones of these layouts.
[0055] It is known from Felix Stopp et al (2021) arXiv:2108.06948v1 [quant-ph] that altering the amplitude A of the RF signal VRF over a given time Δt (also referred to as ramping) may allow trapped particles to leave the trapping device and to enter free space. Similarly, particles coming from free space may be captured by the trapping device in a similar manner. However, in practical applications it may be challenging to time and tune the ramping may be difficult.
[0056] According to examples of the present disclosure, the trap layout may be modified so that trapped particles may leave and enter a trapping device without having to ramp the amplitude of the RF signal.
[0057] Referring now to FIGS. **3**-**7**, illustrating exemplary devices **300**, **400**, **500**, **600**, **700** for trapping one or more particles according to the present disclosure. Same reference signs as already shown and discussed with regard to FIGS. **1**A-**2**C shall be the same and will not be described again in detail, but reference is made to what has been described further above. Again, the devices **300**, **400**, **500**, **600**, **700** may include additional DC electrodes, e.g., for compensating straight fields (e.g., connected to a ground potential or to another compensation potential) and/or for shielding that are not illustrated in the drawings.
[0058] The exemplary devices **300**, **400**, **500**, **600**, **700** show a similar layout as device **100**A shown in FIG. **1**A and device **200**A shown in FIG. **2**A. However, as discussed further above, with regard to FIGS. **1**A-**2**C other layouts for trapping devices are possible (e.g., where the DC electrodes are arranged in a different fashion and/or some DC electrodes being omitted; or with the first RF electrode **110** and the second RF electrode **120** laying in two different planes as shown in FIG. **2**C). The trapping device according to the present disclosure shall not be limited to the specific layout shown in FIGS. **3**-**7** but shall encompass any of such variations.
[0059] Still referring to FIGS. **3**-**7**, the devices **300**, **400**, **500**, **600**, **700** are similar to the devices **100**A, **100**B, **100**C, **200**A, **200**B, **200**C as already described above, in that they have a same or similar center region **170** with the first RF electrode **110**, the second RF electrode **120** and the DC electrodes **130**.sub.1, . . . ,, **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**, **132**, **142** being arranged in a same or similar fashion in the center region **170** In addition, in the ending region **180**, the first RF electrode **110** and the second RF electrode **120** of the devices **300**, **400**, **500**, **600**, **700** (as exemplarily illustrated in FIGS. **3**-**7**) are tapered, or spaced further away from each other (as compared to in the center region) or a combination of both. This may reduce a contribution of the potential generated by the RF electrodes **110** and **120** to the first direction x in the ending region **180**, thereby allowing trapped particles to leave the trapping device more easily and/or allowing incoming particles to enter the trapping device more easily.
[0060] According to some embodiments, in addition to that, a part **152** of the at least one further DC electrode **150** that is located in the ending region **180** may follow the shape of the first RF electrode **110** and the second electrode **120** such that respective distances between the first side **112** of the first RF electrode **110** and the further DC electrode(s) **150** as well as between the first side **122** of the second RF electrode **120** and the further DC electrode(s) **150** in the edge region **180** remain substantially the same as in the center region **170**. While in FIGS. **3**-**7** part **152** of DC electrode **150** is illustrated as part of a single DC electrode **150** extending from the center region **170** to the ending region **180**, it may also possible that part **152** in the ending region **180** is a separate DC electrode or comprises multiple DC electrode segments (as for example, illustrated in FIG. **1**B). In some examples, a width of part **152** is larger than the first distance d.sub.1,c. The part **152** of DC electrode **150** in the ending region **180** as illustrated in FIGS. **3**-**7** is not necessarily required, but is an optional feature and could not be present. In that case, no DC electrode or no

part of a DC electrode may be present between the first RF electrode **110** and the second RF electrode **120** in the ending region **180**.

[0061] While not shown in FIGS. **3-7**, in some examples, an outer side of the ending region **180** of the first RF electrode **110** and the second RF electrode **120** may be aligned with an edge of substrate **160**. In other words, substrate **160** may end (e.g., in the first direction x) where ends **116** and **126** of the first RF electrode **110** and the second RF electrode are located such that ends **116** and **126** are located at an edge of substrate **160**.

[0062] In embodiments, in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, one or both of the following two features is fulfilled: (i) the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120** have a distance d.sub.1 that is greater than the first distance d.sub.1,c; and (ii) the second side **114** of the first RF electrode **110** and the second side **124** of the second RF electrode **120** have a distance d.sub.2 that is less than the second distance d.sub.2,c. The first distance d.sub.1,c and the second distance d.sub.2,c are taken in the center region **170** of the first RF electrode **110** and the second RF electrode **120**. As discussed above, the first distance d.sub.1,c is taken between the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120**. The second distance d.sub.2,c is taken between the second side **112** of the first RF electrode **110** and the second side **122** of the second RF electrode **120**. The first distance, the second distance d.sub.2,c, the distance d.sub.1 and the distance d.sub.2 may be measured along a same direction (such as a direction that is orthogonal to the first direction x, as explained further above).

[0063] Now referring to FIGS. **3** and **5-7**, illustrating exemplary trapping devices **300**, **500**, **600**, **700** according to the present disclosure. In the embodiments exemplarily illustrated in FIGS. **3** and **5-7**, in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, the distance d.sub.1 between the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120** increases towards the end **116** of the first RF electrode **110** and towards the end **126** of the second RF electrode **120** along the first direction X.

[0064] In some embodiments, as exemplarily illustrated in FIG. **5**, the distance d.sub.2 between the second side **114** of the first RF electrode **110** and the second side **124** of the second RF electrode **120** decreases towards the end **116** of the first RF electrode **110** and towards the end **126** of the second RF electrode **120**.

[0065] In other embodiments, as exemplarily illustrated in FIGS. **6** and **7**, the distance d.sub.2 between the second side **114** of the first RF electrode **110** and the second side **124** of the second RF electrode **120** increases towards the end **116** of the first RF electrode **110** and towards the end **126** of the second RF electrode **120**.

[0066] According to further embodiments, as exemplarily illustrated in FIG. **3**, the distance d.sub.2 between the second side **114** of the first RF electrode **110** and the second side **124** of the second RF electrode **120** substantially remains constant towards the end **116** of the first RF electrode **110** and towards the end **126** of the second RF electrode **120**.

[0067] Referring now to FIGS. **4** and **5**, illustrating exemplary trapping devices **400** and **500** according to the present disclosure. In the embodiments exemplarily illustrated in FIGS. **4** and **5**, in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, the distance d.sub.2 between the second side **114** of the first RF electrode **110** and the second side **124** of the second RF electrode **120** decreases towards an end **116** of the first RF electrode **110** and towards an end **126** of the second RF electrode **120**.

[0068] In an embodiment, as exemplarily illustrated in FIG. **5**, the distance d.sub.1 between the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120** increases towards the end **116** of the first RF electrode **110** and towards the end **126** of the second RF electrode **120**.

[0069] In another embodiment, as exemplarily illustrated in FIG. **4**, the distance d.sub.1 between the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120** substantially remains constant towards the end **116** of the first RF electrode **110** and towards the end **126** of the second RF electrode **120**. In yet another embodiment, that is otherwise similar to the embodiment illustrated in FIG. **4**, the distance d.sub.1 between the first side **112** of the first RF electrode **110** and the first side **122** of the second RF electrode **120** decreases towards the end **116** of the first RF electrode **110** and towards the end **126** of the second RF electrode **120**. In this embodiment, the first RF electrode **110** and the second RF electrode **120** has a tapered shape in the ending region **180** (e.g., the decrease in d.sub.2 is greater than the decrease in d.sub.1 in the ending region **180**).

[0070] Referring now to FIGS. **3**, **5**, and **7** illustrating exemplary trapping devices **300**, **500** and **700** according to the present disclosure. In the embodiments exemplarily illustrated in FIGS. **3**, **5** and **7**, in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, the first side **112** of the first RF electrode **110** and the second side **114** of the first RF electrode **110** have a distance d.sub.3 that is less than the third distance d.sub.3,c (where d.sub.3,c being taken in the center region **170** in accordance with what has been described further above). In addition or as an alternative, in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, the first side **122** of the second RF electrode **120** and the second side **124** of the second RF electrode **120** have a distance d.sub.4 that is less than the fourth distance (d.sub.4,c being taken in the center region **170** in accordance with what has been described further above).

[0071] Referring now to FIG. **6** illustrating an exemplary trapping device **600** according to the present disclosure. In the embodiment exemplarily illustrated in FIG. **6**, in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, the first side **112** of the first RF electrode **110** and the second side **114** of the first RF electrode **110** have a distance d.sub.3 that is substantially equal to the third distance d.sub.3,c, and the first side **122** of the second RF electrode **120** and the second side **124** of the second RF electrode **120** have a distance d.sub.4 that is substantially equal to the fourth distance d.sub.4,c (d.sub.3,c and d.sub.4,c being taken in the center region **170** in accordance with what has been described further above).

[0072] Referring now to FIGS. **3-5**, and **7** illustrating exemplary trapping devices **300**, **400**, **500** and **700** according to the present disclosure. In the embodiments exemplarily illustrated in

[0073] FIGS. **3-5** and **7**, in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, the first RF electrode **110** and the second RF electrode **120** each have a tapered shape.

[0074] Referring now to FIGS. **3** and **5-7** illustrating exemplary trapping devices **300**, **500**, **600** and **700** according to the present disclosure. In the embodiments exemplarily illustrated in FIGS. **3** and **5-7**, in the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, the first RF electrode **110** and the second RF electrode **120** are spaced further away from each other as compared to in the center region **170**.

[0075] Although the exemplary illustrated devices **300**, **400**, **500**, **600**, **700** are illustrated with a linearly changing distances d.sub.1, d.sub.2, d.sub.3, and d.sub.4 in the ending region **180**, also different slopes may be possible and the present disclosure shall not be limited to linear slopes. Similarly, although the exemplarily illustrated devices **300**, **400**, **500** and **700** are illustrated with a sharp end **116** of the first RF electrode **110** and a sharp end **126** of the second RF electrode **120**, the ends **116** and **126** do not necessarily have to be sharp so that also different shapes of ends **116** and **126** shall be encompassed by the present disclosure.

[0076] In the following exemplary applications of a trapping device **300**, **400**, **500**, **600**, **700** according to the present disclosure are set forth. For example, since an RF signal being applied to the first RF electrode **110** and the second RF electrode **120** may remain unaltered while particles are leaving or entering the trapping device (such as the amplitude may remain being constant), the trapping device **300**, **400**, **500**, **600**, **700** according to the present disclosure may allow a mode of operation, where some particles are being trapped at the center region **170** of the trapping device **300**, **400**, **500**, **600**, **700** while at the same time other particles may enter or leave at the ending region **180** of the same trapping device **300**, **400**, **500**, **600**, **700**. In other words, the same pair of first RF electrode **110** and second RF electrode **120** can be used for both trapping and loading/de-loading particles in the trapping device **300**, **400**, **500**, **600**, **700**. For trapping device layouts that require an RF ramp for sending particles to free space or re-capturing the particles from free space at the ending region, the RF ramp may not allow to keep another set of particles being trapped by the same device in the center region (but due to the RF ramp, the trapped particles in the center region may be lost).

[0077] For example, the device **300**, **400**, **500**, **600**, **700** for trapping one or more particles according to the present disclosure may be configured to receive one or more particles at the ending region **180** of the first RF electrode **110** and the second RF electrode **120**, and trap the received one or more particles in the center region **170** of the first RF electrode **110** and the second RF electrode **120** using the plurality of DC electrodes **130**.sub.1, . . . **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**. In the same or in other examples, an RF signal being applied to the first RF electrode **110** and the second RF electrode **120** may remain unaltered (such as the amplitude A and the frequency w of the RF signal VRF may remain constant) during the reception of the one or more particles at the ending region **180** of the first RF electrode **110** and the second RF electrode **120**. For example, at least some of the plurality DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**, **132**, **142** may be used to receive (such as capture) the one or more particles. In other examples, additional DC electrodes arranged on the first substrate **160** or second substrate **162** may be used to receive (such as capture) the one or more particles. For example, DC voltage pulses may be applied to the mentioned electrodes for capturing the particles.

[0078] In yet the same or in different examples, the device **300**, **400** **500**, **600**, **700** is configured to accelerate one or more particles from the center region **170** towards the ending region **180** such that the one or more particles leave the ending region **180**. In the same or in other examples, an RF signal being applied to the first RF electrode **110** and the second RF electrode **120** may remain unaltered (such as the amplitude A and the frequency w of the RF signal VRF may remain constant) during the acceleration of the one or more particles at the ending region **180** of the first RF electrode **110** and the second RF electrode **120**. For example, at least some of the plurality DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**, **132**, **142** may be used to accelerate the one or more particles. In other examples, additional electrodes arranged on the first substrate **160** or second substrate **162** may be used to accelerate the one or more particles. In addition to that or as an alternative, electrodes that are not arranged on any of the first and second substrates (not shown) may be used to accelerate the one or more particles.

[0079] Regarding the velocity of incoming particles to be trapped with the device **300**, **400**, **500**, **600**, **700** according to the present disclosure, it will be clear to the person skilled in the art, that the direction of the velocity should point towards the first direction x (into which the first RF electrode **110** and the second RF electrode **120** extend). In addition to that, it will be clear to the skilled person that the kinetic energy of the incoming particles should be higher than the remaining potential in the first direction x in the ending region **180** of the trapping device **300**, **400**, **500**, **600**, **700** and lower than the potentials in first, second and third directions in the center region **170** of the trapping device generated by the first RF electrode **110**, the second RF electrode **120** and the DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**, **132**, **142** for trapping the particles. The direction and kinetic energy of the incoming particles may be controlled by appropriate measures (e.g., using one or more Sikler lenses) which are known to persons skilled in the art.

[0080] FIG. **8** illustrates yet another application for one or more of the trapping devices described in the present disclosure. FIG. **8** illustrates a system **800** including a first device **810** for trapping one or more particles (such as ions, molecules or electrons) and a second device **820** for trapping one or more particles according to an example of the present disclosure.

[0081] Generally, the first trapping device **810** and the second trapping device can be any one of the trapping devices **100**A, **100**B, **100**C, **200**A, **200**B, **200**C, **300**, **400**, **500**, **600**, **700** as described above with reference to FIGS. **1**A-**7**. In some embodiments, one or both of the first trapping device **810** and the second trapping device **820** is a trapping device **300**, **400**, **500**, **600**, **700** as described above with reference to FIGS. **3-7**. In other embodiments, one or both of the first device **810** and second device **820** may be a trapping device as described with regard to FIGS. **1**A-**1**C and a ramp of the RF signal may be used to send the one or more particles from the first device **810** to the second device **820**.

[0082] In some practical applications, one or more particles are loaded from a particle source **840** to a trapping device for performing one or more quantum gate operations (e.g., for quantum computing). Typically, the particle source **840** is located in close proximity or integrated to the trapping device **810**. When loading ions into a trapping device **810**, a typical problem that arises is that some of the particles coming from the particle source **840** may enter the center region **180** of the trapping device **810**, but may not be trapped by the trapping device **810**. This may cause a contamination of surfaces of at least one of the first RF electrode, the second RF electrode **110**, **120** and the plurality of DC electrodes **130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**, **132**, **142**, which in turn may cause stray fields that may influence or deteriorate the trapping potential of the electrodes.

[0083] According to the present disclosure, a first trapping device **810** may be used to trap one or more particles from a particle source **840**, and a separate second trapping device **820** may be used for performing one or more quantum gate operations on the one or more particles.

[0084] The first device **810** may be configured to provide the one or more trapped particles to the second device **820** for further processing. In some examples, a valve element **830** may be present between the first device **810** and the second device **820** that may only open when one or more particles are provided from the first device **810** to the second device **820**.

[0085] The second device **820** may further be configured to trap the one or more particles and subsequently move the one or more particles from a first position of the second device **820** to a second position of the second device **820**, where a quantum gate operation unit **850** may be present.

[0086] According to embodiments, the first device **810** and the second device **820** may be spaced away from each other by a distance that is larger than the second distance d.sub.2,c. According to examples, quantum gate operation unit **850** may include applying one or more of a laser light and a microwave to the trapped one or more particles.

[0087] In one example, the particle source **840** may include an ablation target and one or more ionization lasers. In the same or in another example, the particle source **840** may include a magneto-optical trap.

[0088] The valve element **830** may include a tube and/or a valve for controlling transport of the one or more particles from the first device **810** to the second device **820**.

[0089] Separating the first device **810** (which is used for receiving the particles from the particle source **840**) from the second device **820** (which is used for performing the quantum gate operation) may hinder or decrease a surface contamination of the electrodes in the second device **820**. For the first device **810** surface contaminations may be acceptable, for example, in case the first device **810** is used only for providing the particles to the second trap **820** and not for performing quantum gate operations. The second device **820** may be devoid of any additional particle source (except from the particles that are provided to the second device **820** from the first device **810**).

[0090] FIG. **9** illustrates a method **900** for providing one or more particles (such as ions) to a device for trapping one or more particles. For example, method **900** may be used in conjunction with a system **800** as exemplarily described above with regard to FIG. **8**.

[0091] The method **900** comprises trapping **910**, by a first device **810**, at least one particle provided by a particle source **840**. The method **900** further comprises providing **920**, by the first device, the at least one particle to a second device **820**. The method **900** further comprises receiving **930**, by the second device **820**, the at least one particle. The method **900** optionally further comprises moving the at least one received particle from a first position of the second device **820** to a second position of the second device **820**. The method **900** further comprises performing **950** one or more quantum gate operations on the at least one particle.

[0092] The examples described herein provide:

[0093] Example 1. A device (**200**A, **200**B, **200**C, **300**, **400**, **500**, **600**, **700**) for trapping one or more particles, the device comprising: a first radio-frequency (RF) electrode (**110**), a second RF electrode (**120**), and a plurality of direct current (DC) electrodes (**130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**, **132**, **142**); wherein the first RF electrode (**110**) and the second RF electrode (**120**) extend in a first direction (x), wherein the first RF electrode (**110**) comprises a first side (**112**) and a second side (**114**), wherein the second side (**114**) of the first RF electrode (**110**) is arranged opposite to the first side (**112**) of the first RF electrode (**110**), wherein the second RF electrode (**120**) comprises a first side (**122**) and a second side (**124**), wherein the second side (**124**) of the second RF electrodes (**120**) is arranged opposite the first side (**122**) of the second RF electrode (**120**), wherein the first side (**112**) of the first RF electrode (**110**) faces towards the first side (**122**) of the second RF electrode (**120**), wherein, in a center region (**170**) of the first RF electrode (**110**) and the second RF electrode (**120**), the first side (**112**) of the first RF electrode (**110**) and the first side (**122**) of the second RF electrode (**120**) have a first distance (d.sub.1,c) with respect to each other, wherein, in the center region (**170**) of the first RF electrode (**110**) and the second RF electrode (**120**), the second side (**112**) of the first RF electrode (**110**) and the second side (**122**) of the second RF electrode (**120**) have a second distance (d.sub.2,c) with respect to each other, and wherein, in an ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**): the first side (**112**) of the first RF electrode (**110**) and the first side (**122**) of the second RF electrode (**120**) have a distance (d.sub.1) that is greater than the first distance (d.sub.1,c); and/or the second side (**114**) of the first RF electrode (**110**) and the second side (**124**) of the second RF electrode (**120**) have a distance (d.sub.2) that is less than the second distance (d.sub.2,c).

[0094] Example 2. The device (**200**A, **200**B, **300**, **400**, **500**, **600**, **700**) of example 1, wherein the first RF electrode (**110**) and the second electrode (**120**) are arranged on a first substrate (**160**) such that the first RF electrode and the second RF electrode are arranged within a common plane.

[0095] Example 3. The device (**200**C, **300**, **400**, **500**, **600**, **700**) of example 1, wherein the first RF electrode (**110**) is arranged on a first substrate (**160**) and the second RF electrode (**120**) is arranged on a second substrate (**162**) such that the first RF electrode and the second RF electrode are arranged in two separate planes.

[0096] Example 4. The device (**200**A, **200**B, **200**C, **300**, **400**, **500**, **600**, **700**) of any of the preceding examples, wherein, in the center region (**170**) of the first RF electrode (**110**) and the second RF electrode (**120**), the first distance (d.sub.1,c) between the first side (**112**) of the first RF electrode (**110**) and the first side (**122**) of the second RF electrode (**120**) is substantially constant.

[0097] Example 5. The device (**200**A, **200**B, **200**C, **300**, **400**, **500**, **600**, **700**) of any of the preceding examples, wherein, in the center region (**170**) of the first RF electrode (**110**) and the second RF electrode (**120**), the second distance (d.sub.2,c) between the second side (**114**) of the first RF electrode (**110**) and the second side (**124**) of the second RF electrode (**120**) is substantially constant.

[0098] Example 6. The device (**200**A, **200**B, **200**C, **300**, **400**, **500**, **600**, **700**) of any of the preceding examples, wherein the distance (d.sub.1), the distance (d.sub.2), the first distance (d.sub.1,c) and the second distance (d.sub.2,c) are measured in a direction that is orthogonal to the first direction (x).

[0099] Example 7. The device (**200**A, **200**B, **200**C, **300**, **400**, **500**, **600**, **700**) of any of the preceding examples, wherein the plurality of DC electrodes (**130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**) is arranged adjacent to the first RF electrode (**110**) and the second RF electrode (**120**) in the center region (**170**) of the first RF electrode (**110**) and the second RF electrode (**120**).

[0100] Example 8. The device (**200**A, **200**B, **200**C, **300**, **400**, **500**, **600**, **700**) of example 7, wherein a first set (**130**.sub.1, . . . , **130**.sub.n) of the plurality of DC electrodes (**130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**) is arranged adjacent the second side (**114**) of the first RF electrode (**110**), and wherein a second set (**140**.sub.1, . . . , **140**.sub.n) of the plurality of DC electrodes (**130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**) is arranged adjacent the second side (**124**) of the second RF electrode (**120**).
[0101] Example 9. The device (**200**A, **200**B, **200**C, **300**, **400**, **500**, **600**, **700**) of example 8, wherein at least a further DC electrode (**150**) of the plurality of DC electrodes (**130**.sub.1, **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**) is arranged between the first side (**112**) of the first RF electrode (**110**) and the first side (**122**) of the second RF electrode (**120**).
[0102] Example 10. The device (**200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) of any of the preceding examples, wherein, in the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), the distance (d.sub.1) between the first side (**112**) of the first RF electrode (**110**) and the first side (**122**) of the second RF electrode (**120**) increases towards an end (**116**) of the first RF electrode (**110**) and towards an end (**126**) of the second RF electrode (**120**).
[0103] Example 11. The device (**200**A, **200**B, **200**C, **500**) of example 10, wherein the distance (d.sub.2) between the second side (**114**) of the first RF electrode (**110**) and the second side (**124**) of the second RF electrode (**120**) decreases towards the end (**116**) of the first RF electrode (**110**) and towards the end (**126**) of the second RF electrode (**120**).
[0104] Example 12. The device (**200**A, **200**B, **200**C, **600**, **700**) of example 10, wherein the distance (d.sub.2) between the second side (**114**) of the first RF electrode (**110**) and the second side (**124**) of the second RF electrode (**120**) increases towards the end (**116**) of the first RF electrode (**110**) and towards the end (**126**) of the second RF electrode (**120**).
[0105] Example 13. The device (**200**A, **200**B, **200**C, **300**) of example 10, wherein the distance (d2) between the second side (**114**) of the first RF electrode (**110**) and the second side (**124**) of the second RF electrode (**120**) substantially remains constant towards the end (**116**) of the first RF electrode (**110**) and towards the end (**126**) of the second RF electrode (**120**).
[0106] Example 14. The device (**200**A, **200**B, **200**C, **400**, **500**) of any of examples 1 to 9, wherein, in the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), the distance (d.sub.2) between the second side (**114**) of the first RF electrode (**110**) and the second side (**124**) of the second RF electrode (**120**) decreases towards an end (**116**) of the first RF electrode (**110**) and towards an end (**126**) of the second RF electrode (**120**).
[0107] Example 15. The device (**200**A, **200**B, **200**C, **500**) of example 14, wherein the distance (d.sub.1) between the first side (**112**) of the first RF electrode (**110**) and the first side (**122**) of the second RF electrode (**120**) increases towards the end (**116**) of the first RF electrode (**110**) and towards the end (**126**) of the second RF electrode (**120**).
[0108] Example 16. The device (**200**A, **200**B, **200**C, **400**) of example 14, wherein the distance (d.sub.1) between the first side (**112**) of the first RF electrode (**110**) and the first side (**122**) of the second RF electrode (**120** substantially remains constant towards the end (**116**) of the first RF electrode (**110**) and towards the end (**126**) of the second RF electrode (**120**).
[0109] Example 17. The device (**200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) of any one of the preceding examples, wherein, in the center region (**170**) of the first RF electrode (**110**) and the second RF electrode (**120**), the first side (**112**) of the first RF electrode (**110**) and the second side (**114**) of the first RF electrode (**110**) have a third distance (d.sub.3,c) with respect to each other and the first side (**122**) of the second RF electrode (**120**) and the second side (**124**) of the second RF electrode have a fourth distance (d.sub.4,c) with respect to each other, wherein, in the center region (**170**), the third distance (d.sub.3,c) is substantially constant, and the fourth distance (d.sub.4,c) is substantially constant.
[0110] Example 18. The device (**200**A, **200**B, **200**C, **300**, **500**, **700**) of example 17, wherein, in the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), the first side

(**112**) of the first RF electrode (**110**) and the second side (**114**) of the first RF electrode (**110**) have a distance (d.sub.3) that is less than the third distance (d.sub.3,c).
[0111] Example 19. The device (**200**A, **200**B, **200**C, **300**, **500**, **700**) of example 18, wherein, in the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), the first side (**122**) of the second RF electrode (**120**) and the second side (**124**) of the second RF electrode (**120**) have a distance (d.sub.4) that is less than the fourth distance (d.sub.4,c).
[0112] Example 20. The device (**200**A, **200**B, **200**C, **600**) of example 17, wherein, in the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), the first side (**112**) of the first RF electrode (**110**) and the second side (**114**) of the first RF electrode (**110**) have a distance (d.sub.3) that is substantially equal to the third distance (d.sub.3,c), and the first side (**122**) of the second RF electrode (**120**) and the second side (**124**) of the second RF electrode (**120**) have a distance (d.sub.4) that is substantially equal to the fourth distance (d.sub.4,c).
[0113] Example 21. The device (**200**A, **200**B, **200**C, **300**, **400**, **500**, **700**) of any one of examples 1 to 19, wherein, in the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), the first RF electrode (**110**) and the second RF electrode (**120**) each have a tapered shape.
[0114] Example 22. The device (**200**A, **200**B, **200**C, **300**, **500**, **500**, **700**) of any one of examples 1 to 19, wherein, in the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), the first RF electrode (**110**) and the second RF electrode (**120**) are spaced further away from each other as compared to in the center region (**170**)
[0115] Example 23. The device (**200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) of any of the preceding examples, wherein the device is configured to: receive one or more particles at the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), and trap the received one or more particles in the center region (**170**) of the first RF electrode (**110**) and the second RF electrode (**120**) using the plurality of DC electrodes (**130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**).
[0116] Example 24. The device (**200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) of example 23, wherein during the reception of the one or more particles at the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), an amplitude of an RF signal being applied to the first RF electrode (**110**) and the second RF electrode (**120**) remains unaltered.
[0117] Example 25. The device (**200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) of any of the preceding examples, wherein the device is configured to: accelerate one or more particles from the center region (**170**) towards the ending region (**180**) such that the one or more particles leave the ending region.
[0118] Example 26. The device (**200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) of example 25, wherein at least some of the plurality DC electrodes (**130**.sub.1, . . . , **130**.sub.n, **140**.sub.1, . . . , **140**.sub.n, **150**) are used to accelerate the one or more particles.
[0119] Example 27. The device (**200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) of example 26, wherein during the acceleration of the one or more particles at the ending region (**180**) of the first RF electrode (**110**) and the second RF electrode (**120**), an amplitude of an RF signal being applied to the first RF electrode (**110**) and the second RF electrode (**120**) remains unaltered.
[0120] Example 28. The device (**200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) of example 1, wherein, in the ending region (**180**) one or more of the plurality of DC electrodes are arranged between the first side (**112**) of the first RF electrode (**110**) and the first side (**122**) of the second RF electrode (**120**), wherein a width of the one or more of the plurality of DC electrodes is larger than the first distance (d.sub.1,c).
[0121] Example 29. A system (**800**) comprising: a first device (**810**; **100**A, **100**B, **100**C, **200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) for trapping one or more particles; a second device (**820**; **100**A, **100**B, **100**C, **200**A, **200**B, **200**C, **300**, **500**, **600**, **700**) for trapping one or more particles; and wherein the first device is configured to trap (**910**) at least one particle provided by a particle

source (**840**), wherein the second device (**820**) is configured to: receive (**930**) the at least one particle from the first device (**810**), and perform (**950**) one or more quantum gate operations on the at least one particle.
[0122] Example 30. A method (**900**) of providing one or more particles to a device (**820**) for trapping one or more particles, the method comprising: trapping (**910**), by a first device, at least one particle provided by a particle source (**840**), providing (**920**), by the first device, the at least one particle to a second device (**820**), receiving (**930**), by the second device, the at least one particle, and performing (**950**), by the second device, one or more quantum gate operations on the at least one particle.
[0123] Although specific examples have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific examples shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific examples discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.
[0124] The expression "and/or" should be interpreted to cover all possible conjunctive and disjunctive combinations, unless expressly noted otherwise. For example, the expression "A and/or B" should be interpreted to mean A but not B, B but not A, or both A and B. The expression "at least one of" should be interpreted in the same manner as "and/or", unless expressly noted otherwise. For example, the expression "at least one of A and B" should be interpreted to mean A but not B, B but not A, or both A and B.
[0125] It should be noted that the methods and devices including its preferred embodiments as outlined in the present document may be used stand-alone or in combination with the other methods and devices disclosed in this document. In addition, the features outlined in the context of a device are also applicable to a corresponding method, and vice versa. Furthermore, all aspects of the methods and devices outlined in the present document may be arbitrarily combined. In particular, the features of the claims may be combined with one another in an arbitrary manner.
[0126] It should be noted that the description and drawings merely illustrate the principles of the proposed methods and systems. Those skilled in the art will be able to implement various arrangements that, although not explicitly described or shown herein, embody the principles of the invention and are included within its spirit and scope. Furthermore, all examples and embodiments outlined in the present document are principally intended expressly to be only for explanatory purposes to help the reader in understanding the principles of the proposed methods and systems. Furthermore, all statements herein providing principles, aspects, and embodiments of the invention, as well as specific examples thereof, are intended to encompass equivalents thereof.

Claims

1. A device for trapping one or more particles, the device comprising: a first radio-frequency (RF) electrode; a second RF electrode; and a plurality of direct current electrodes, wherein the first RF electrode and the second RF electrode extend in a first direction, wherein the first RF electrode comprises a first side and a second side, wherein the second side of the first RF electrode is arranged opposite to the first side of the first RF electrode, wherein the second RF electrode comprises a first side and a second side, wherein the second side of the second RF electrode is arranged opposite the first side of the second RF electrode, wherein the first side of the first RF electrode faces towards the first side of the second RF electrode, wherein in a center region of the first RF electrode and the second RF electrode, the first side of the first RF electrode and the first side of the second RF electrode have a first distance with respect to each other, and the second side of the first RF electrode and the second side of the second RF electrode have a second distance with respect to each other, and wherein in an ending region of the first RF electrode and the second

RF electrode: the first side of the first RF electrode and the first side of the second RF electrode have a distance that is greater than the first distance; and/or the second side of the first RF electrode and the second side of the second RF electrode have a distance that is less than the second distance.

2. The device of claim 1, wherein the first RF electrode and the second RF electrode are arranged on a first substrate such that the first RF electrode and the second RF electrode are arranged within a common plane.

3. The device of claim 1, wherein the first RF electrode is arranged on a first substrate and the second RF electrode is arranged on a second substrate such that the first RF electrode and the second RF electrode are arranged in two separate planes.

4. The device of claim 1, wherein in the center region of the first RF electrode and the second RF electrode, the first distance between the first side of the first RF electrode and the first side of the second RF electrode is substantially constant, and wherein in the center region of the first RF electrode and the second RF electrode, the second distance between the second side of the first RF electrode and the second side of the second RF electrode is substantially constant.

5. The device of claim 1, wherein each of the distances is measured in a direction that is orthogonal to the first direction.

6. The device of claim 1, wherein in the ending region of the first RF electrode and the second RF electrode, the distance between the first side of the first RF electrode and the first side of the second RF electrode increases towards an end of the first RF electrode and towards an end of the second RF electrode.

7. The device of claim 6, wherein the distance between the second side of the first RF electrode and the second side of the second RF electrode decreases towards the end of the first RF electrode and towards the end of the second RF electrode.

8. The device of claim 6, wherein the distance between the second side of the first RF electrode and the second side of the second RF electrode increases towards the end of the first RF electrode and towards the end of the second RF electrode.

9. The device of claim 6, wherein the distance between the second side of the first RF electrode and the second side of the second RF electrode substantially remains constant towards the end of the first RF electrode and towards the end of the second RF electrode.

10. The device of claim 1, wherein in the ending region of the first RF electrode and the second RF electrode, the distance between the second side of the first RF electrode and the second side of the second RF electrode decreases towards an end of the first RF electrode and towards an end of the second RF electrode.

11. The device of claim 10, wherein the distance between the first side of the first RF electrode and the first side of the second RF electrode increases towards the end of the first RF electrode and towards the end of the second RF electrode.

12. The device of claim 10, wherein the distance between the first side of the first RF electrode and the first side of the second RF electrode substantially remains constant towards the end of the first RF electrode and towards the end of the second RF electrode.

13. The device of claim 1, wherein in the center region of the first RF electrode and the second RF electrode, the first side of the first RF electrode and the second side of the first RF electrode have a third distance with respect to each other and the first side of the second RF electrode and the second side of the second RF electrode have a fourth distance with respect to each other, and wherein in the center region, the third distance is substantially constant and the fourth distance is substantially constant.

14. The device of claim 13, wherein in the ending region of the first RF electrode and the second RF electrode, the first side of the first RF electrode and the second side of the first RF electrode have a distance that is less than the third distance.

15. The device of claim 14, wherein in the ending region of the first RF electrode and the second RF electrode, the first side of the second RF electrode and the second side of the second RF electrode have a distance that is less than the fourth distance.

16. The device of claim 15, wherein in the ending region of the first RF electrode and the second RF electrode, the first side of the first RF electrode and the second side of the first RF electrode have a distance that is substantially equal to the third distance and the first side of the second RF electrode and the second side of the second RF electrode have a distance that is substantially equal to the fourth distance.

17. The device of claim 1, wherein in the ending region of the first RF electrode and the second RF electrode, the first RF electrode and the second RF electrode each have a tapered shape.

18. The device of claim 1, wherein in the ending region of the first RF electrode and the second RF electrode, the first RF electrode and the second RF electrode are spaced further away from each other as compared to in the center region.

19. The device of claim 1, wherein the device is configured to: receive one or more particles at the ending region of the first RF electrode and the second RF electrode; and trap the received one or more particles in the center region of the first RF electrode and the second RF electrode using the plurality of DC electrodes.

20. The device of claim 19, wherein during the reception of the one or more particles at the ending region of the first RF electrode and the second RF electrode, an amplitude of an RF signal being applied to the first RF electrode and the second RF electrode remains unaltered.

21. The device of claim 1, wherein the device is configured to: accelerate one or more particles from the center region towards the ending region such that the one or more particles leave the ending region.

22. The device of claim 21, wherein at least some of the plurality DC electrodes are used to accelerate the one or more particles.

23. The device of claim 22, wherein during the acceleration of the one or more particles at the ending region of the first RF electrode and the second RF electrode, an amplitude of an RF signal being applied to the first RF electrode and the second RF electrode remains unaltered.

24. The device of claim 1, wherein in the ending region, one or more of the plurality of DC electrodes is arranged between the first side of the first RF electrode and the first side of the second RF electrode, and wherein a width of the one or more of the plurality of DC electrodes is larger than the first distance.

25. A system, comprising: a first device configured to trap one or more particles; and a second device configured to trap one or more particles, wherein the first device is configured to trap at least one particle provided by a particle source, wherein the second device is configured to: receive the at least one particle from the first device; and perform one or more quantum gate operations on the at least one particle.

26. A method of providing one or more particles to a device for trapping one or more particles, the method comprising: trapping, by a first device, at least one particle provided by a particle source; providing, by the first device, the at least one particle to a second device; receiving, by the second device, the at least one particle; and performing, by the second device, one or more quantum gate operations on the at least one particle.
